

ميد برمجة 1 دفعة 21

Answer the following multiple-choice questions:

1. The concept used in the statement `int a = (int) 56.75` is called (3 marks)
 - a. Illegal assignment
 - b. Type casting
 - c. Integer overflow
 - d. Rounding
2. Which of the following is a numerical data type? (3 marks)
 - a. short
 - b. String
 - c. char
 - d. all of the above
3. Which of the following is a wrong statement when declaring a variable? (3 marks)
 - a. `double payment;`
 - b. `short $age11;`
 - c. `int 5cm;`
 - d. `long cm2mile;`
4. What will be displayed by the following switch statement: (3 marks)

```
int b = 5;
switch(b) {
    case 1: System.out.print(b);
    case 3: System.out.print(b);
    case 5: System.out.print('b');
    case 6: System.out.println("Five");
    break;
}
```

 - a. b
 - b. bbbFive
 - c. bFive
 - d. Five
5. If `x=1` and `y=2`, which of the following expression evaluates to *true*? (4 marks)
 - a. `(x > 0) && (y > 3)`
 - b. `(x < 0) || (y >= 1)`
 - c. `((x < 0) || (y == 1))`
 - d. `!(x > 0) && (y > 3)`
6. The following expression `int x = 3+4*3+2;` results in `x` having the value: (4 marks)
 - a. 11
 - b. 8
 - c. 17
 - d. None of the above

Part II (30 points)

Q1. (20 points) Translate the following algorithm into Java code:

CLO-2

Definition: kWh is a unit of energy equal to one kilowatt of power sustained for one hour
Saudi Electricity Tariffs (Rates) for three categories of consumption

Consumption categories (Kwh) / Month		(1) Residential (Halalah / kwh)	(2) Commercial (Halalah / kwh)	(3) Agricultural & Charities (Halalah / kwh)
A	1-6000	18	20	16
B	More than 6000	30	30	20

Note: An additional tax charge of 15% is added to the bill.

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Algorithm:

- 1- Read (input) the class type (1 for Residential or 2 for Commercial 3 for Agricultural & Charities)
- 2- Read (input) consumption in kWh/month (integer)
- 3- Evaluate the rate based on consumption, consumption category and class type and set the respective rate
- 4- Calculate the value of consumption in Riyals using the formula: $VoC = (\text{consumption} \times \text{Rate value}) + 10$
- 5- Calculate the Value Added Tax (VAT) using the formula: $\text{Value of Consumption in Riyals} \times 0.15$
- 6- Calculate the total required payment amount using the formula: $\text{Total} = \text{Value of Consumption} + \text{VAT}$
- 7- Finally display the message "The consumption category is [A or B] and the Total is [amount] SAR"

Sample input/output

Sample #1

Enter class type: 1 for Residential or 2 for Commercial 3 for Agricultural & Charities

1

Enter Number Of Consumption in kWh/month

2000

The consumption category is A and the Total is 414.0 SAR

Sample #2

Enter class type: 1 for Residential or 2 for Commercial 3 for Agricultural & Charities

2

Enter Number Of Consumption in kWh/month

5500

The consumption category is A and the Total is 1263.0 SAR

Sample #3

Enter class type: 1 for Residential or 2 for Commercial 3 for Agricultural & Charities

2

Enter Number Of Consumption in kWh/month

13000

The consumption category is B and the Total is 4485.0 SAR

Solution:

```
Scanner input=new Scanner(System.in);
```

```
System.out.println("Enter class type: 1 for Residential or 2 for Commercial 3 for  
Agricultural & Charities ");  
  
int classType = input.nextInt();  
System.out.println("Enter Number Of Consumption in kWh/month");  
int numberOfConsumption = input.nextInt();  
char consumptionCategory = 'N';  
int rate = 0;  
double valueOfConsumption = 0;  
double VAT = 0;  
double total = 0;  
if(numberOfConsumption>= 1 && numberOfConsumption <= 6000) {  
    consumptionCategory='A';  
    if(classType==1) {  
        rate=18;  
    }  
    else if(classType==2){  
        rate=20;  
    }  
    else if(classType==3){  
        rate=16;  
    }  
    else {  
        System.out.println("Invalid class type");  
    }  
}  
else if(numberOfConsumption>6000) {  
    consumptionCategory = 'B';  
    if(classType==1 || classType==2 ) {  
        rate=30;  
    }  
    else if(classType==3) {  
        rate=20;  
    }  
    else {  
        System.out.println("Invalid class type");  
    }  
}  
valueOfConsumption = (numberOfConsumption* rate)/100;  
VAT = valueOfConsumption *0.15;  
total = valueOfConsumption+VAT;  
System.out.println("The consumption category is " + consumptionCategory + " and the  
Total is " + total + " SAR");
```

Q2. (10 points)

CLO-12

Write the output of the following expressions :

1) `int A = 65 ;`
`System.out.println( "A" + 'A' + A );`
`System.out.println ( 'A' + A + "A" );`
`System.out.println( A + "A" + 'A' );`
`System.out.println( 'A' + "A" + A );`
(4 points)

AA65
130A
65AA
AA65

2) `char num = '5';`
`switch(num){`
 `case '1' :System.out.print(num);`
 `case '5' :System.out.print(num);`
 `case '2' :System.out.print('2');`
 `case '4' :System.out.println("num");`
`}`
(2 points)

Output: 52num

3) `int X=1, sum=0;`
`while(X++ <5) {`
 `if( X %2 == 0)`
 `System.out.print( X + " + " );`
 `sum += X ;`
 `++X;`
`}`
`System.out.println(X);`
`double avg = (double) sum/X ;`
`System.out.println("Avg = " + avg) ;`
(4 points)

Output: 2 + 4 + 6
Avg = 1.0

Part III: (50 points)

Q1. (10 points)

CLO-13

The following program has been written by a new Java programmer to read ONLY one sentence (group of words) from the user. The program is intended to capitalize the whole sentence as shown in sample output. However, the program has syntax, runtime, and logic errors. Please mark [✓] for all lines that have no error and rewrite the ones that have error in the correction table.

Hint: there are 6 lines that have errors, where one of them is already corrected.

1	package midtermExam202;
2	import java.util.Scanner;
3	public class MidtermExam202
4	{
5	public static void main(String[] args) {
6	Scanner read = new Scanner(System.in)
7	System.out.print("Please enter the sentence which is required to be capitalized: ");
8	String line = read.next();
9	int x = 1;
10	String capitalizedSentence = "";
11	while (x <= line.length())
12	{
13	if (line.charAt(x) <= 'a' && line.charAt(x) >= 'z') {
14	char c = (char) (line.charAt(x) - 97 + 65);
15	capitalizedSentence += c;}
16	else {
17	capitalizedSentence += line.charAt(x);}
18	x+;
19	}
20	System.out.println(capitalizedSentence) };

Note: In the table below we will keep non-highlighted line as it is, this means students only focus

Note: In the table below we will keep non-highlighted line as it is, this means students only focus on lines from 6 to 15.

No.	Write your correction code for each line that has error or mark [✓] if no error
1	✓
2	✓
3	✓
4	✓
5	✓
6	Scanner read = new Scanner(System.in);
7	✓
8	String line = read.nextLine();
9	int x = 0;
10	✓
11	while (x < line.length())
12	✓
13	if (line.charAt(x) >= 'a' && line.charAt(x) <= 'z') {
14	✓
15	✓
16	✓
17	✓

Good Luck ☺

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18	x++;
19	✓
20	✓

Q2. (40 points)

Write a program that reads a string from the user and determines the following **type** of characters present in the string:

1. vowel : (a or A), (e or E), (i or I), (o or O), (u or U)
2. consonant : other (non-vowel) letters of the English Alphabet like b, D, Y, z, etc.
3. other : not alphabet letters like %, ?, \$, @, 5, +, etc.

Your output should print the following

1. vowel letters present in the **String**
2. count of each vowel letter present in the string.
3. Consonants present in the **string**
4. non-alphabet characters present in the **string**

[Hint : a non-alphabet character will not be between 'a' and 'z' or between 'A' and 'Z']

CLO-7, 8

Sample run:

```
Please enter a string : Welcome@KAU!!
Vowels entered : eoeAU
(a or A) : 1
(e or E) : 2
(i or I) : 0
(o or O) : 1
(u or U) : 1
Consonants entered : WlcmK
Other characters entered : @!!
```

- | | |
|---|-------------|
| 1. Correct printing of vowel letters present in String | (10 points) |
| 2. Correct calculation and printing of count of each vowel present in the string. | (10 points) |
| 3. Correct printing of consonants present in the string | (10 points) |
| 4. Correct printing of non-alphabet characters present in the string | (10 points) |

Sample code given as a guide only. Student solutions may vary considerably. Please use own judgement when awarding marks

```
Scanner input = new Scanner(System.in);

System.out.print("Please enter a string : ");
String sentence = input.next();

int len = sentence.length();

int pos = 0;
```

```

        if ((c == 'a' || c == 'A') ||
            (c == 'e' || c == 'E') ||
            (c == 'i' || c == 'I') ||
            (c == 'o' || c == 'O') ||
            (c == 'u' || c == 'U')) System.out.print(c);

        // count type of characters
        if (c == 'a' || c == 'A') a++;
        else if (c == 'e' || c == 'E') e++;
        else if (c == 'i' || c == 'I') i++;
        else if (c == 'o' || c == 'O') o++;
        else if (c == 'u' || c == 'U') u++;

        pos++;
    }

    System.out.println();
    System.out.println("(a or A) : " + a);
    System.out.println("(e or E) : " + e);
    System.out.println("(i or I) : " + i);
    System.out.println("(o or O) : " + o);
    System.out.println("(u or U) : " + u);

    pos = 0;
    System.out.print("Consonants entered : ");
    while(pos < len) {
        char c = sentence.charAt(pos);
        // print consonants

        if (((c >= 'a' && c <= 'z') || (c >= 'A' && (c <= 'Z'))) &&
            (c != 'a' && c != 'A') &&
            (c != 'e' && c != 'E') &&
            (c != 'i' && c != 'I') &&
            (c != 'o' && c != 'O') &&
            (c != 'u' && c != 'U')) System.out.print(c);

        pos++;
    }
    System.out.println();

    pos = 0;
    System.out.print("Other characters entered : ");
    while(pos < len) {
        char c = sentence.charAt(pos);
        // print other characters

        if (!(c >= 'a' && c <= 'z') && !(c >= 'A' && (c <= 'Z')))
            System.out.print(c);

        pos++;
    }
    System.out.println();

```